

5.2.49

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Question

The eigenvalues of the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}$$

are λ_1, λ_2 and λ_3 . Find the value of

$$\lambda_1 \lambda_2 \lambda_3 (\lambda_1 + \lambda_2 + \lambda_3).$$

Solution

We know that for any square matrix:

$$\lambda_1 \lambda_2 \lambda_3 = \det \mathbf{A}, \quad \lambda_1 + \lambda_2 + \lambda_3 = \text{trace}(\mathbf{A}). \quad (1)$$

$$\text{trace}(\mathbf{A}) = 3 + 5 + 3 = 11. \quad (2)$$

$$\det \mathbf{A} = 3(15 - 1) - (-1)(-3 + 1) + (1 - 5) \quad (3)$$

$$= 3 \cdot 14 - 2 - 4 \quad (4)$$

$$= 36. \quad (5)$$

Hence

$$\lambda_1 \lambda_2 \lambda_3 (\lambda_1 + \lambda_2 + \lambda_3) = \det \mathbf{A} \cdot \text{trace}(\mathbf{A}) = 36 \cdot 11 = 396. \quad (6)$$

```
#include <stdio.h>

int main() {
    int A[3][3] = {
        {3, -1, 1},
        {-1, 5, -1},
        {1, -1, 3}
    };

    int trace = 0;
    for (int i = 0; i < 3; i++) {
        trace += A[i][i];
    }

    // Compute determinant using cofactor expansion along first
    row
}
```

C Code

```
int det = 0;
for (int j = 0; j < 3; j++) {
    int minor = A[1][(j+1)%3]*A[2][(j+2)%3] - A[1][(j+2)%3]*A[2][(j+1)%3];
    if (j % 2 == 0)
        det += A[0][j] * minor;
    else
        det -= A[0][j] * minor;
}

int result = det * trace;

printf(Trace = %d\n, trace);
printf(Determinant = %d\n, det);
printf(lambda1*lambda2*lambda3*(lambda1+lambda2+lambda3) = %d\n, result);

return 0;
}
```

Python Direct

```
import numpy as np

# Define the matrix
A = np.array([[3, -1, 1],
              [-1, 5, -1],
              [1, -1, 3]])

trace = np.trace(A)

# Compute determinant
det = np.linalg.det(A)

result = det * trace

print(Trace =, trace)
print(Determinant =, det)
print(lambda1*lambda2*lambda3*(lambda1+lambda2+lambda3) =, result
)
```

```
import ctypes

# Load the shared library
lib = ctypes.CDLL('./libmatrix_calc.so')

# Tell ctypes the return type of the function
lib.compute_result.restype = ctypes.c_int

# Call the function
result = lib.compute_result()

print(lambda1*lambda2*lambda3*(lambda1+lambda2+lambda3) =, result
)
```